# Case 03 · Data protection impact assessment (DPIA): Roll Transparency Log

*An additional document born from the POC's key finding: a transparent log of the voter roll is also information about people. Structure follows a common DPIA template. An educational document, not legal advice.*

## Processing

- **Collected per change:** voter pseudonym (HMAC of elector number), action and reason code, booth, date, officer pseudonym, hash of the evidence document.
- **Who sees what:** the public sees counts by district, week and code; the person sees their own lines in full, with proof; the Commission sees the roll as today; witnesses see hashes and heads only.
- **Retention:** leaves are kept forever, which is the point of the log; therefore they contain nothing that identifies a person without the key.

## Necessity and proportionality

Without a reasoned record of every change, nobody can know what happened to 130 million names or appeal in time. The less intrusive-looking alternative, a one-off list of the removed with names (as ordered for Bihar in 2025), is in fact more intrusive and weaker.

## Risks

| # | Risk | Likelihood | Severity |
|---|---|---|---|
| P1 | Re-identification from booth and date (POC: almost every booth under 5 changes) | High | High |
| P2 | Pseudonym dictionary if the key leaks | Low | Very high |
| P3 | An R-DEAD line reaches the family first | Medium | Medium |
| P4 | Profiling of communities by movement | Medium | High |
| P5 | Self-check abused to look up someone else | Medium | Medium |
| P6 | Officer identified via pseudonym | Low | Medium |

## Measures

- **P1:** public feed by district and week; cells under 10 shown as "<10"; booth and date only in the personal proof.
- **P2:** key in an HSM with a second HSM elsewhere; no export; planned key rotation.
- **P3:** R-DEAD is not sent by SMS; shown only in self-check or kiosk.
- **P4:** no religion, caste, language or surname field anywhere; coarse codes.
- **P5:** number plus date of birth; rate limit; only the pseudonym's own lines.
- **P6:** officer pseudonyms rotate per revision; the link is opened only in an appeal.

## Residual risk

P1 high → low (if the threshold holds across combinations); P2 very high → low; P3 medium → low; P4 high → medium (needs external review); P5 medium → medium; P6 medium → low.

## Conclusion

The processing is justified and proportionate, provided the public view is aggregate. The largest design change was born here: transparency is for the person; the public gets only sums.